# Hierarchical Position Encoding for Structure-Sensitive Semantic Dense Retrieval

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## 1. Problem Definition

Every major embedding model in production — BGE-M3, Qwen3-Embedding, E5-Mistral, Nomic Embed — encodes token position as a single integer via Rotary Position Embedding (RoPE). RoPE computes attention scores as a function of relative sequence distance only: Attention(A,B) ~ f(|pos_A - pos_B|). Depth in a document tree, parent section identity, and sibling index are architecturally invisible.

This produces a systematic retrieval failure. The sentence "Revenue grew 40%" embedded under the heading "## Q3 Performance > Revenue" and the same sentence under "## Q3 Risks > Revenue" produce near-identical embedding vectors in every current production model (cosine similarity ≈ 0.97). A RAG system querying "what are the Q3 risks?" may retrieve the performance-framed chunk as equally relevant as the risk-framed chunk, causing the downstream language model to generate a confidently wrong answer.

The failure is not limited to markdown. It affects financial tables (row/column coordinate lost), JSON configuration files (key-path nesting invisible), and legal contracts (clause hierarchy ignored). BEIR and MTEB — the benchmarks all production embedding models optimize for — contain zero structurally-adversarial test pairs, meaning this failure mode was never measured and no model ever optimized against it.

**Root Cause:** RoPE encodes one number per token — sequence position — and discards every other dimension of where a token lives in a document. The problem is not data size or model scale. It is the inductive bias: the assumption that linear sequence distance is sufficient to represent semantic proximity.

## 2. Introduction

### *2.1 The RAG Landscape and Retrieval Bottleneck*

The global Retrieval-Augmented Generation (RAG) market was valued at $2.33 billion in 2025 and is projected to reach $81.51 billion by 2035 at a CAGR of 42.7%. Document retrieval is the dominant application, accounting for 32.4% of all RAG revenue. Despite this growth, naive RAG systems achieve only 10–40% success rates in enterprise environments — with retrieval failure identified as the primary bottleneck, not generation failure. 71% of organizations use GenAI regularly yet only 17% report meaningful business impact. The gap is retrieval quality.

### *2.2 Embedding Models and Their Architectural Limitation*

Dense retrieval systems convert documents and queries into fixed-size vector representations using transformer-based embedding models trained via contrastive learning. The dominant positional mechanism — RoPE — rotates query and key vectors by an angle proportional to sequence position, producing attention scores that reflect linear token distance. Every production model inherits this design. The pipeline is: Document → Chunk → Embed (RoPE) → Vector DB → Query Match → LLM. The embedding step is the architectural bottleneck for structured document retrieval.

### *2.3 HiSE — The Proposed Solution*

This project proposes HiSE (Hierarchical Structure-aware Embeddings), incorporating HiSA (Hierarchical Structural Addressing): a novel positional encoding mechanism extending RoPE from 1-dimensional sequence position to 4-dimensional structural address. The formula changes from $\theta = \theta_1 \times m$ (standard RoPE) to $\theta = \theta_1 m + \theta_2 d + \theta_3 p + \theta_4 s$ (HiSA), where m = sequence position, d = depth in document tree, p = parent section hash, s = sibling index. Only 96 new scalar parameters are added — a surgically small change with potentially large structural effect.

## 3. Literature Review

### *3.1 Positional Encoding in Transformers*

Rotary Position Embedding (RoPE) was introduced by Su et al. (2024) and is now the standard positional mechanism in all major LLMs and embedding models — Llama, Qwen, Mistral, DeepSeek, BGE-M3. RoPE encodes position as a single integer and produces relative distance as the attention signal. Vaswani et al. (2017) established the original sinusoidal positional encoding in "Attention Is All You Need", which RoPE subsequently improved upon for efficiency and length generalization. Neither mechanism has any concept of hierarchical structural position.

### *3.2 Structure-Aware Encoding Prior Work*

Multiple works have explored structure-aware encoding for specific domains. TUTA (Wang et al., KDD 2021) applied bi-dimensional tree coordinates to table cells for classification tasks — not retrieval. DOM-LM (Deng et al., 2022) injected depth, parent index, and sibling index as additive position embeddings into BERT for HTML — HTML-only, additive not rotational, no dense retrieval training. Struc-EMB (Liu et al., Georgia Tech, 2025) tested structural context integration in LLM-based embedders via prompt concatenation — a prompt-level workaround rather than an architectural fix. RethinkRoPE (Liu et al., 2025) provided a mathematical blueprint for extending RoPE to N dimensions using Lie group theory — for vision and spatial domains only. None of these works trained a contrastive retrieval model with structural hard negatives validated on a structural sensitivity benchmark.

**Table 1: Literature Review — Prior Work and Research Gap**

| Paper | Year | Contribution | Limitation / Gap |
|---|---|---|---|
| Su et al. — RoFormer (RoPE) | 2024 | Introduced rotary position embedding (RoPE) — rotating Q and K vectors by angle proportional to sequence position | Encodes only 1D linear position; structurally blind to heading path, table coordinates, JSON key-path |
| Vaswani et al. — Attention Is All You Need | 2017 | Foundational transformer with sinusoidal position encodings; established the Q·K attention framework | No structural awareness; position = flat integer sequence only |

| Paper | Year | Contribution | Limitation / Gap |
|---|---|---|---|
| Wang et al. (Microsoft) — TUTA | 2021 | Tree-based transformer with bi-dimensional coordinates for table cell understanding | Classification tasks only — never trained for dense retrieval or RAG |
| Deng et al. — DOM-LM | 2022 | Injects depth, parent index, sibling index as additive PE into BERT for HTML document understanding | HTML-only, additive not rotational, not trained for dense retrieval |
| Liu et al. (Georgia Tech) — Struc-EMB | 2025 | Investigates sequential and parallel structural context integration in LLM-based embedders | Prompt-level workaround — no positional encoding architecture change, no structural hard negatives |
| Liu et al. — RethinkRoPE | 2025 | Mathematical blueprint for extending RoPE to N-dimensional positional embedding using Lie group theory | Proposed for vision and spatial domains — not applied to document hierarchy or text retrieval |
| Thakur et al. — BEIR | 2021 | Heterogeneous benchmark for zero-shot evaluation of information retrieval models | Zero structurally-adversarial test pairs — cannot measure structural discrimination failure |
| Muennighoff et al. — MTEB | 2023 | Massive text embedding benchmark covering 56 tasks across retrieval, clustering, classification | Zero structurally-adversarial test pairs — does not expose the structural blindness failure mode |

## 4. Motivation

HiSE is motivated by four converging observations:

- Research Gap in Positional Encoding: Existing positional encoding methods exclusively model sequential relationships. Hierarchical structural position — depth, parent path, sibling index — remains unexplored for dense retrieval, despite being the primary semantic signal in enterprise documents.
- Structure Carries Meaning: The same sentence under '## Q3 Performance' and '## Q3 Risks' means opposite things. Headings, table row/column headers, and JSON key-paths encode semantics that current embedding models discard entirely.

- Benchmark Blindness: BEIR and MTEB have zero structurally-adversarial test pairs. The failure was never measured, so no model ever optimized against it. StructEval-Bench v1 is the first benchmark to expose this gap — its creation is itself a research contribution.
- Production Consequence: Enterprise RAG systems run over hierarchically structured documents daily — financial reports, legal contracts, technical specifications, configuration files. The higher the structural complexity, the more confidently wrong the retrieval becomes. Naive RAG's 10–40% success rate is directly traceable to this architectural blind spot.

HiSE addresses the root cause — not by adding more data, more parameters, or better prompts — but by correcting the inductive bias at the positional encoding layer where the failure originates.

## 5. Objectives

- G1 — Design and validate HiSA: a novel positional encoding mechanism extending RoPE from 1D sequence position to 4D structural address (sequence position, depth, parent identity, sibling index).
- G2 — Train HiSE-Embed: a contrastive retrieval model under 300M parameters with HiSA replacing standard RoPE, trained with structurally-adversarial hard negatives.
- G3 — Build StructEval-Bench v1: the first retrieval benchmark with 300–500 structurally-adversarial pairs across markdown heading hierarchies, tabular row-column structures, and nested JSON key-paths.
- O1 — Achieve statistically significant improvement in Structural Discrimination Score (SDS) on StructEval-Bench v1 versus a same-size same-data same-compute standard RoPE baseline.
- O2 — Maintain nDCG@10 on BEIR/MTEB subsets within ±2% margin of error — structural gains must not degrade general retrieval.
- O3 — Run controlled ablation isolating HiSA as the experimental variable: same backbone, same data, same compute, only positional encoding changes.

- O4 — Release HiSE-Embed, StructEval-Bench v1, and HiSA implementation open-source under asl-research/hise (GitHub + HuggingFace).
- O5 — Submit research paper to ICNLP 2027 (IEEE Xplore) by November 30, 2026.

## 6. Methodology Used

### *6.1 HiSA — Architectural Mechanism*

HiSA replaces the single-integer position input to RoPE with a 4-dimensional structural address tuple (m, d, p, s). Each dimension receives its own learned rotation frequency:

$$\textbf{Standard RoPE: } \theta = \theta_1 \times m$$

$$\textbf{HiSA: } \theta = \theta_1 m + \theta_2 d + \theta_3 p + \theta_4 s$$

Where m = sequence position, d = depth in document tree (H1=1, H2=2, H3=3 / JSON nesting level), p = parent section hash (fixed-size integer), s = sibling index under same parent. The degenerate case guarantee: when d=0, p=0, s=0, HiSA reduces exactly to standard RoPE — ensuring no performance regression on plain-text benchmarks.

### *6.2 Training Strategy — Structurally-Adversarial Hard Negatives*

HiSE-Embed is trained via contrastive learning (InfoNCE loss) with two categories of negatives: (a) standard semantic hard negatives — high-cosine-similarity documents that are semantically different, and (b) structurally-adversarial hard negatives — documents with near-identical surface text but structurally-opposed ground-truth relevance. Category (b) is the novel training signal that forces the model to actively use the structural dimensions rather than ignoring them during optimization.

### *6.3 Document Parsers*

Three format-specific parsers extract structural addresses from input documents: (a) Markdown parser — extracts heading hierarchy depth and path using Python-Markdown AST; (b) Table parser — extracts row and column header identities and cell coordinates using pandas; (c) JSON parser — extracts key-path depth and full key-path hash using recursive traversal. All three formats share the same (m, d, p, s) tuple interface, demonstrating HiSA's format-agnosticism.

### *6.4 Ablation Design*

The ablation study holds constant: SLM backbone architecture, training corpus, compute budget, contrastive loss function, batch size, learning rate. The single variable is the positional encoding mechanism — standard RoPE versus HiSA. This design isolates HiSA as the causal factor for any observed improvement in structural discrimination scores.

## 7. Development Plan

**Table 2: Phase-wise Development Plan**

| Phase | Timeline | Deliverable | Success Criterion |
|---|---|---|---|
| 1 | Months 1–2 | Failure-mode survey — quantitative proof that current models fail on structurally-adversarial pairs | cosine sim ≥ 0.90 between structurally-opposed pairs on BGE-M3 / Qwen3-Embedding |
| 2 | Month 2–3 | HiSA implementation — replace standard RoPE with 4D structural address in small SLM backbone | Unit tests pass: degenerate case (d=0,p=0,s=0) produces identical output to standard RoPE |
| 3 | Months 3–4 | StructEval-Bench v1 — 300–500 structurally-adversarial pairs, SDS and RIR metrics defined | Benchmark released on HuggingFace Datasets with annotation guidelines |
| 4 | Months 4–5 | HiSE-Embed training + ablation — same backbone, data, compute; only positional encoding changes | SDS improvement statistically significant ($p < 0.05$) vs standard RoPE baseline |
| 5 | Month 6 | Evaluation on BEIR/MTEB + paper writing + open-source release | nDCG@10 within ±2% of baseline; ICNLP 2027 submission November 30, 2026 |

Hardware: Primary training on ASL Server 'Glory' — AMD Ryzen 9 7900X, NVIDIA RTX 5070 Ti 16GB VRAM, 32GB DDR5 RAM, Ubuntu 24.04 LTS. Development on ROG Strix 'Void' — Intel i7 13th Gen, NVIDIA RTX 4070 8GB VRAM.

Software: Python 3.11+, PyTorch 2.x, CUDA 12.8, HuggingFace Transformers, Sentence-Transformers, FAISS, Weights & Biases, Docker.

## 8. Conclusion

HiSE addresses a fundamental and previously unmeasured architectural limitation in dense retrieval embedding models: the inability to encode hierarchical structural position. The contribution is not a domain-specific fix or a prompt-engineering workaround. It is a targeted modification to the positional encoding mechanism itself — validated through controlled ablation and a purpose-built benchmark.

Three independent research artifacts constitute the project's contributions: HiSA (the positional encoding mechanism), HiSE-Embed (the trained retrieval model), and StructEval-Bench v1 (the benchmark). Each is independently verifiable and independently publishable. Together they represent the first complete system for structure-sensitive dense retrieval — from mechanism to model to evaluation.

HiSE does not improve an existing model by training on more data. It corrects the inductive bias: replacing the assumption that position in a sequence is sufficient to represent position in a document. The target publication is ICNLP 2027 (IEEE Xplore), with all artifacts released open-source under the asl-research/hise GitHub organization.

## 9. References

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